Applying Gaussian mixture models to the Na-O plane to separate multiple populations in globular clusters

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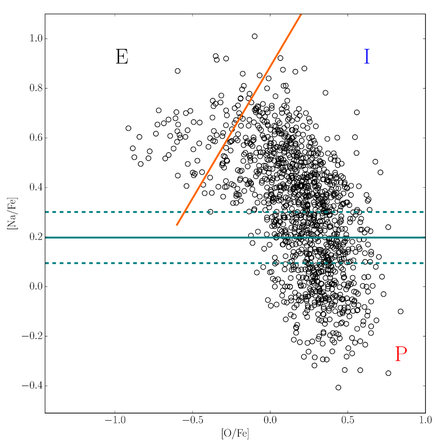
Motivation

- Stellar populations in globular clusters (GCs) are usually classified as primordial **(P)** and enhanced **(I or E)** populations on an individual basis based on a cluster specific minimum [Na/Fe] value.
- Stars below this minimum [Na/Fe] threshold are considered primordial, and stars above the threshold are labeled enhanced.
- The number of primordial and enhanced stars in GCs has important implications for what is responsible for the Na-O anti-correlation and our understanding of cluster evolution.
- The available [Na/Fe] and [O/Fe] abundances for 19 GCs from the Carretta et al. and Gratton et al. (2006 -2015) papers provide a rich and homogeneous dataset.
- We present a new method of classifying the populations in GCs using multi-dimensional Gaussian Mixture Models (GMM) in the Na-O plane based on this dataset in its entirety, rather than on a cluster-by-cluster basis.
- The GCs analyzed in this study are listed on the right with their Mv, [Fe/H], half light radius in arcminutes, and central concentration.
- The structural parameters were taken from the Harris Catalog 2010 version.

NGC	Mv	[Fe/H]	rh	c
104	-9.42	-0.74	3.17	2.07
288	-6.75	-1.24	2.23	0.99
1904	-7.86	-1.55	0.65	1.70
2808	-9.39	-1.13	0.80	1.56
3201	-7.45	-1.50	3.10	1.29
4590	-7.37	-2.24	1.51	1.41
5904	-8.81	-1.35	1.77	1.73
6121	-7.19	-1.20	4.33	1.65
6171	-7.12	-1.06	1.73	1.53
6218	-7.31	-1.31	1.77	1.34
6254	-7.48	-1.56	1.95	1.38
6388	-9.41	-0.41	0.52	1.75
6397	-6.64	-1.99	2.90	2.50
6441	-9.63	-0.36	0.57	1.74
6752	-7.73	-1.56	1.91	2.50
6809	-7.57	-1.97	2.83	0.93
6838	-5.61	-0.82	1.67	1.15
7078	-9.19	-2.33	1.00	2.29
7099	-7.45	-2.34	1.03	2.50

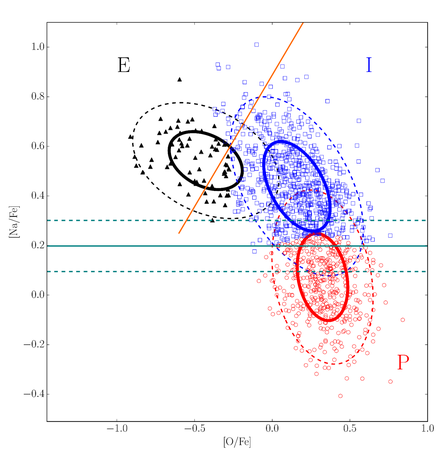
Cluster Data



- Plot of [Na/Fe] vs [O/Fe] abundances for stars in all 19 clusters.
- The solid green line marks the average of the individual [Na/Fe] thresholds to used separate the primordial and enhanced stars in each cluster. The dashed lines are the 1-sigma levels
- The orange line is the boundary used to further separate the enhanced population into intermediate **(I)** and extreme **(E)** stars. This line is defined as [O/Na] < -0.9 by Carretta et al. 2009 VII.
- We have also labeled the regions of the plot with their respective classifications based on the previous method of separating the populations.
- It is this entire sample of stars that we will use in constructing our GMM.

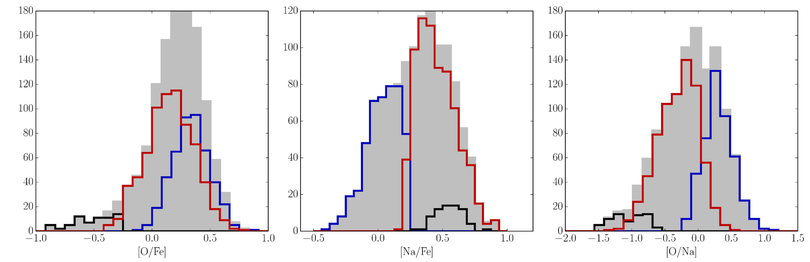
Constructing the GMM

- A Gaussian mixture model is composed of **N** Gaussian components that are each characterized by a mixing weight, covariance matrix, and mean values in each dimension of the data being considered. In this case the dimensions are [O/Fe], [Na/Fe], and [O/Na].
- Each of the **N** components may be used as a class label for unsupervised clustering where no previous labels are known, as is the case with the multiple populations in these globular clusters.
- To determine these model parameters we implemented the expectation maximization algorithm available in the Python scikit-learn GMM package with a **3** component model.
- One component for the **primordial (P)**, **indeterminate (I)**, and **extreme (E)** stars, respectively.
- The final computed GMM allows us to assign a probability that a given star was generated by each of these components given its [O/Fe], [Na/Fe], and [O/Na] abundances.
- A star is assigned to the component that had the highest probability of being responsible for generating that star given its measured abundances.



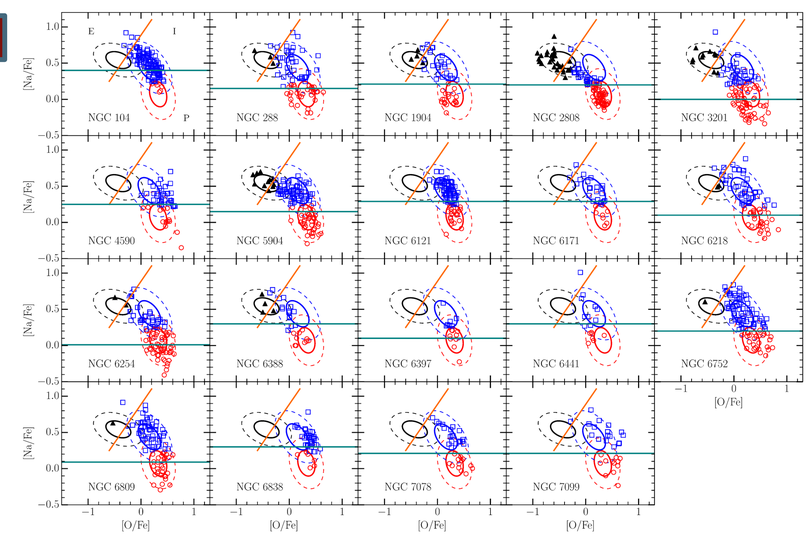
- Here we show the results of classifying the stars based on the GMM constructed from the entire sample.
- The solid and dashed ellipses mark the 1 and 2 sigma levels of each of the Gaussian components: **red ellipses** for the **P component**, **blue ellipses** for the **I component**, and **black ellipses** for the **E component**.
- Stars have been color coded to match the component to which they were assigned.
- The green and orange lines mark the abundance boundaries discussed in the previous plot.
- A comparison between the position of these lines and the classification of stars according to the GMM provides an initial illustration of the differences between these techniques.

Population Distributions



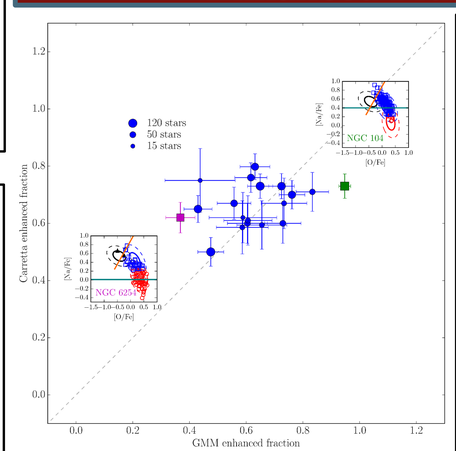
- Plot of the [O/Fe], [Na/Fe], and [O/Na] distributions of the entire sample are in gray.
- The individual distributions of the **P**, **I**, and **E** stars (as labeled by the GMM) are over plotted following the same color scheme as the previous plot.
- This plot illustrates how the various populations separate in each of the abundance measurements.

Individual Clusters



- In this plot we recreate the [Na/Fe] vs [O/Fe] plot from Carretta et al. 2009 VII.
- The stars in each panel are plotted with a color and shape according to how they were classified by the GMM previously discussed.
- The ellipses once again illustrate the 1 and 2 sigma levels of each component.
- Additionally, each panel has a green horizontal line. This line marks the [Na/Fe] abundance used to separate a given cluster based on the method described in Carretta el at. 2009 VII.

Enhanced Fraction (EF)



- The enhanced fraction in each cluster is defined as:
  EF = (I + E) / (P + I + E)
- This plot is a comparison of the enhanced fraction as determined by the Carretta method vs the enhanced fraction determined by the GMM classification.
- The size of the points are determined by the total number of stars in a given cluster.
- The insets show the Na vs O plots for NGC 104 and NGC 6254, which have the maximum and minimum GMM enhanced fraction, respectively.
- These clusters highlight the diversity of EF values allowed by the GMM, while the Carretta method places them at very similar EF values.

References:

Buitinck et al., 2013 (sklearn)
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Carretta, E., Bragaglia, A., Gratton, R.G., Lucatello, S., & Momany, Y. 2007a,A&A, 464, 927
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Harris Catalog 2010 Version
Vanderplas et al, proc. of CIDU, pp. 47-54, 2012 (astroML examples)

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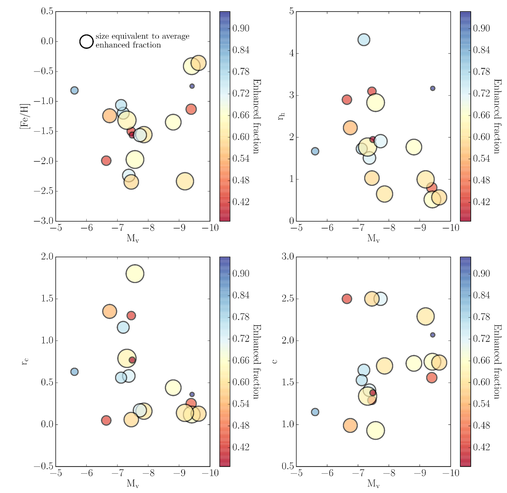
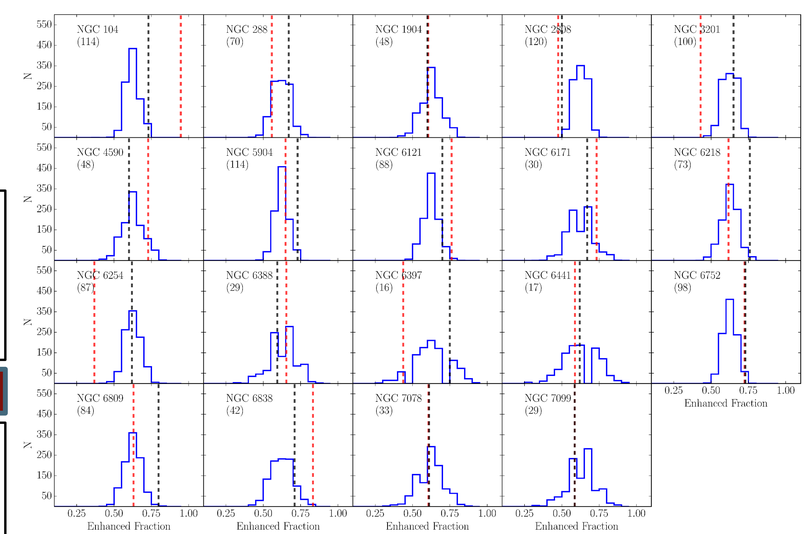
EF vs Cluster parameters



- Plot of various individual GC parameters vs Mv.
- Each circle represents one of the 19 GCs from the sample.
- The circles are color coded and sized according to their EF.
- Clusters with an EF closer to the mean EF are larger than those that are highly deviant.
- We do not see any correlations in EF with these cluster parameters.

Simulated EF Distributions

- Once the GMM is determined it is possible to generate artificial samples of data from the model.
- For each cluster we drew a random artificial sample from the GMM with a number of stars equal to what is actually observed in the data.
- The EF was calculated for this artificial sample and then the process was repeated 500 times.
- These histograms show the distributions of EF values of these 500 iterations.
- The red dashed lines mark the actual EF determined by the GMM.
- The black dashed lines mark the EF as calculated by Carretta el al. 2009 VII.



Conclusions

- The method we present here attempts to use a large homogeneous sample of abundances to model the underlying distribution responsible for the Na-O anti-correlation seen in all GCs with GMMs.
- This method differs from that adopted in previous investigations in which the threshold between primordial and enhanced populations is defined on a cluster-by-cluster basis.
- From our comparison with other work, we see that this method results in a wider range of EF values ~0.3 – 0.6, while the Carretta method gives values of ~0.65, for all clusters.
- It also appears that the clusters with fewer stars in their samples have better agreement between the two methods.
- The simulated EF distributions illustrate that some clusters (NGC 104, NGC 6254) show EF values that differ significantly from what we would expect from the calculated GMM, suggesting an underlying diversity among the GCs. We also found that removing these clusters when calculating the GMM did not drastically change the final calculated model.
- This suggests that the GMM model could be improved in order to better model the behavior of the entire GC population to potentially reduce these large deviations.
- We do not find any correlation in the difference between the mean simulated EF and the GMM calculated value with any cluster parameters ([Fe/H],Mv,rh,c).